Class 27: Reciprocal Space – 1: Introduction to Reciprocal Space

Many properties of solid materials stem from the fact that they have periodic internal structures. Electronic properties are no exception. Why do electronic properties of materials vary from one material to another? One may even be tempted to ask, "Is there a difference in the electrons present in different materials leading to the differences in their electronic properties?" Even within the same material why is graphite conducting in one direction and insulating in another?

In response to the above questions, it is relevant to note that electrons as such are the same in all materials. The difference in electronic properties is therefore not a result of differences in the electrons present in the materials but rather due to 'other differences' between the materials. In crystalline materials the feature that has a significant impact on the electronic properties, is the crystal structure of the specific material. It is therefore of interest to understand the periodic structure of crystalline materials, which is the focus of the present class. While crystal structures are discussed from high school onwards, in this discussion, we will revisit some of the concepts and expand our understanding of periodic structures. We will also familiarize ourselves with a concept referred to as 'Reciprocal space' that is very useful in describing periodic structures. This concept is not very intuitive at first glance, but is very powerful in capturing key features of periodic structures, and hence very useful in understanding the impact of periodicity on electronic properties.

As we saw in the earlier classes, the wave vector $k = \frac{2\pi}{\lambda}$, has the dimensions L^{-1}. We will now examine how we can represent crystallographic information in the same framework as the wave vector information.

Crystal structure has information about crystal directions, axis, unit vectors, which are usually presented in the context of real space where the quantities have the dimensions of L. We will now define a reciprocal space where we will represent the same crystal structure information within a different framework, where the dimensions of the quantities are L^{-1}. This will enable us to more easily relate the crystal structure to the waves of electrons travelling through it, since they are presented in the same framework. Reciprocal space is credited to Ewald, whose work in the 1920s laid the groundwork for this concept.

In the next couple of classes we will look at the reciprocal lattice as an independent entity and then link it back to the models we have examined so far.

In real space we use the unit vectors $\overrightarrow{a_1}$, $\overrightarrow{a_2}$, and $\overrightarrow{a_3}$. We will now define unit vectors for the space that we will call the reciprocal space. It is important to note that the unit vectors for reciprocal space are defined based on our convenience – or rather that they are choices we make. So at first the selection of the unit vectors of the reciprocal lattice seems arbitrary. However, they are deliberately defined in the manner that we will see, because it then gives the corresponding reciprocal space some useful properties and enables interesting relationships with real space.

We will define the unit vectors $\overrightarrow{b_1}$, $\overrightarrow{b_2}$, and $\overrightarrow{b_3}$ in reciprocal space, which relate to the real space vectors $\overrightarrow{a_1}$, $\overrightarrow{a_2}$, and $\overrightarrow{a_3}$, in a specific manner, as shown below.

$$\overrightarrow{b_1} = \frac{\overrightarrow{a_2} \times \overrightarrow{a_3}}{V}$$

$$\overrightarrow{b_2} = \frac{\overrightarrow{a_3} \times \overrightarrow{a_1}}{V}$$

$$\overrightarrow{b_3} = \frac{\overrightarrow{a_1} \times \overrightarrow{a_2}}{V}$$

Where the product designated by '×' is the vector cross product, and V is the volume of the unit cell in real space.

It is important to note that the above definition is something that we are enforcing, since it leads to useful results later on. The calculations and discussions that follow, simply enforce the above definitions.

Let us examine the consequence of the above definition. Let us consider a general triclinic cell in real space. In view of it being triclinic, the three crystal unit vectors need not be equal in length, nor do the angles of the unit cell have to be equal. Therefore, the unit cell in real space is defined as:

$$\overrightarrow{a_1} \neq \overrightarrow{a_2} \neq \overrightarrow{a_3}$$

And

$$\alpha \neq \beta \neq \gamma$$

As an aside, it is important to note that the relation '≠' is used to indicate 'not necessarily equal to', implying, for example, that a cubic cell is a subset of the triclinic cell.

Figure 27.1 below shows a triclinic cell.

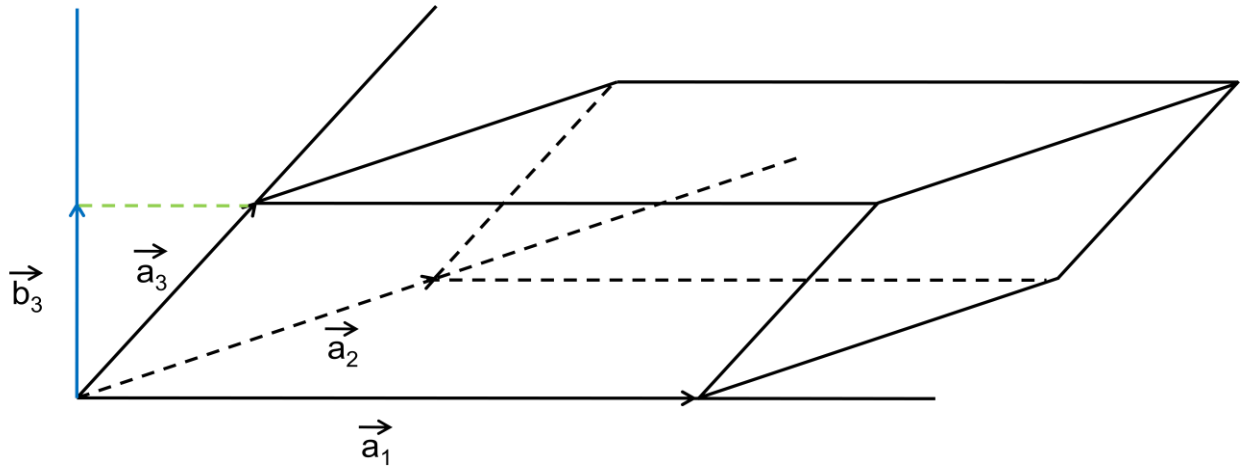


Figure 27.1: A triclinic cell showing the unit vectors $\overrightarrow{a_1}$, $\overrightarrow{a_2}$, and $\overrightarrow{a_3}$. A unit vector of reciprocal space, $\overrightarrow{b_3}$, is also shown on the figure to indicate how it relates to the real space vectors.

Since

$$\overrightarrow{b_3} = \frac{\overrightarrow{a_1} \times \overrightarrow{a_2}}{V}$$

The reciprocal lattice vector $\overrightarrow{b_3}$ is therefore perpendicular to the real lattice vectors $\overrightarrow{a_1}$, and $\overrightarrow{a_2}$, and the plane defined by $\overrightarrow{a_1}$, and $\overrightarrow{a_2}$.

Based on standard vector mathematics, $\overrightarrow{a_1} \times \overrightarrow{a_2}$ is the area of the parallelogram at the base of the triclinic cell shown in Figure 27.1 above, and is the numerator for the equation for $\overrightarrow{b_3}$.

The volume of the triclinic unit cell, which is the denominator for the equation for $\overrightarrow{b_3}$, is simply the product of the area of the base of the unit cell, with the height of the unit cell. Since $\overrightarrow{b_3}$ is perpendicular to the plane defined by $\overrightarrow{a_1}$, and $\overrightarrow{a_2}$, the height of the unit cell is simply the projection of $\overrightarrow{a_3}$ on $\overrightarrow{b_3}$

Therefore

$$|\overrightarrow{b_3}| = \frac{Area\ of\ base\ of\ unit\ cell}{Area\ of\ base\ of\ unit\ cell * Height\ of\ unit\ cell}$$

Simplifying,

$$|\overrightarrow{b_3}| = \frac{1}{Height\ of\ unit\ cell}$$

Since the height of the unit cell represents the distance between nearest plane parallel to the basal plane, it is essentially the spacing between (001) planes in the real space.

Therefore:

$$|\overrightarrow{b_3}| = \frac{1}{d_{001}}$$

Similarly,

$$|\overrightarrow{b_1}| = \frac{1}{d_{100}}$$

And

$$|\overrightarrow{b_2}| = \frac{1}{d_{010}}$$

These are directly a result of how we have defined the relationship between the real space and reciprocal space unit vectors.

Also, due to the definitions,

$$\overrightarrow{b_3}.\overrightarrow{a_2} = 0$$

$$\overrightarrow{b_3}.\overrightarrow{a_1} = 0$$

And

$$\overrightarrow{b_3}.\overrightarrow{a_3} = \frac{1}{Projection\ of\ \overrightarrow{a_3}\ on\ \overrightarrow{b_3}} * Projection\ of\ \overrightarrow{a_3}\ on\ \overrightarrow{b_3} = 1$$

In general

$$\overrightarrow{b_i}.\overrightarrow{a_j} = 0, if\ i \neq j$$

$$= 1, if\ i = j$$

We have looked at the specific case of the unit vectors going to make up the real space and reciprocal space and how they relate to each other. In view of how the reciprocal space is defined, we are able to generalize further. It turns out that if $\overrightarrow{H_{hkl}}$ is a vector in reciprocal space, then it is perpendicular to the plane (hkl) of the real space, and

$$\left|\overrightarrow{H_{hkl}}\right| = \frac{1}{d_{hkl}}$$

Consider Figure 27.2 below which shows the unit vectors of real space, the plane (hkl), and a vector designated as $\overrightarrow{H_{hkl}}$

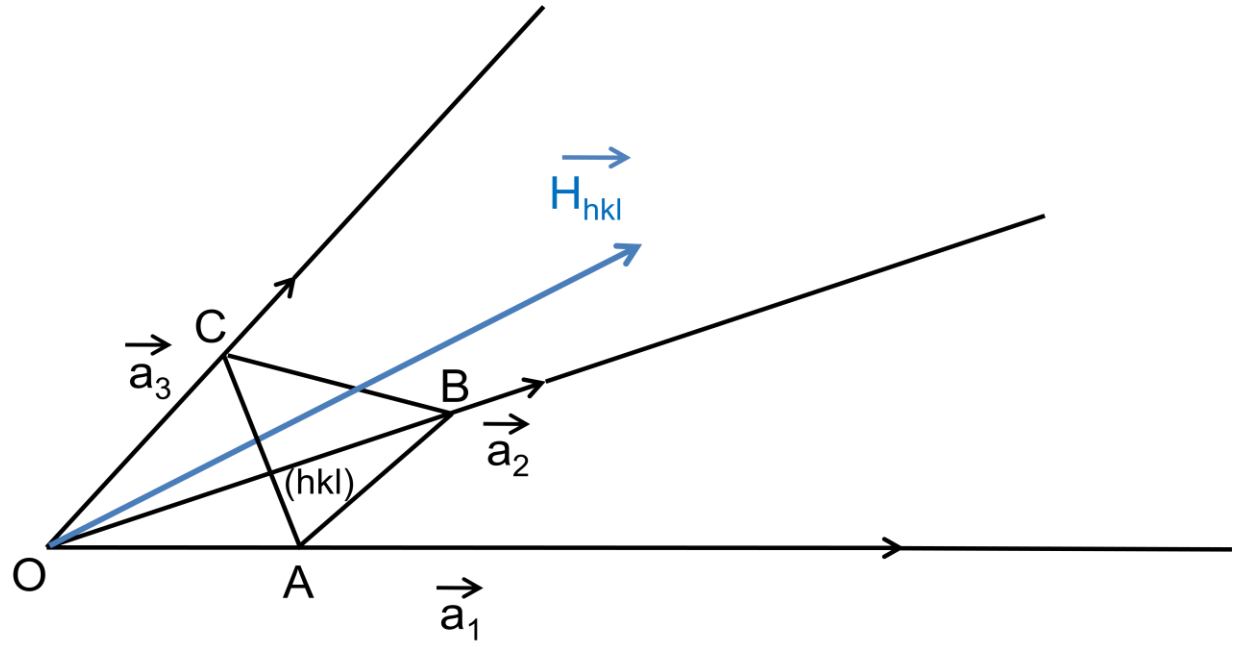


Figure 27.2: The plane (hkl), and a vector designated as $\overrightarrow{H_{hkl}}$

Purely based on conventional nomenclature for vectors, we have

$$\overrightarrow{H_{hkl}} = h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}$$

Where $h, k,$ and l, are integers. We will not associate any other significance to these integers at this time, except to say that they coincide with the miller indices of the plane (hkl) of real space.

Based on the definition of the plane (hkl), the intercept of this plane with the real space axis, occurs at $\frac{\overrightarrow{a_1}}{h}$, $\frac{\overrightarrow{a_2}}{k}$, and $\frac{\overrightarrow{a_3}}{l}$, and are the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$, respectively.

Since

$$\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

Therefore, rearranging and substituting, we get

$$\overrightarrow{AB} = \frac{\overrightarrow{a_2}}{k} - \frac{\overrightarrow{a_1}}{h}$$

This implies,

$$\overrightarrow{H_{hkl}}.\overrightarrow{AB} = \left(h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}\right).\left(\frac{\overrightarrow{a_2}}{k} - \frac{\overrightarrow{a_1}}{h}\right) = -1 + 1 = 0$$

Therefore, $\overrightarrow{H_{hkl}}$ is perpendicular to $\overrightarrow{AB}$. By a similar analysis, it can be shown that $\overrightarrow{H_{hkl}}$ is perpendicular to $\overrightarrow{BC}$, as well as to $\overrightarrow{CA}$. Since any two of the vectors $\overrightarrow{AB}$, $\overrightarrow{BC}$, and $\overrightarrow{CA}$ define the plane ABC, which is also the plane (hkl), we have effectively shown that any reciprocal lattice vector $\overrightarrow{H_{hkl}}$ is perpendicular to a real lattice plane whose miller indices match $h, k,$ and l, i.e the real lattice plane (hkl). Therefore we have now been able to relate the direction of a reciprocal lattice vector to the physical orientation of a real lattice plane.

Let us look at the magnitude of $\overrightarrow{H_{hkl}}$, and examine what it means.

Consider a unit vector along $\overrightarrow{H_{hkl}}$, which we will designate as $\hat{n}$, it is defined by:

$$\hat{n} = \frac{\overrightarrow{H_{hkl}}}{\left|\overrightarrow{H_{hkl}}\right|}$$

In real space, the meaning of defining a plane (hkl) as one which intercepts that respective unit cell vectors at $\frac{\overrightarrow{a_1}}{h}$, $\frac{\overrightarrow{a_2}}{k}$, and $\frac{\overrightarrow{a_3}}{l}$, is that one of the nearest parallel planes of this family, passes through the origin. Therefore d_{hkl}, is simply the shortest distance between the origin and the plane (hkl), or the distance of the origin from the plane (hkl), along the perpendicular to the plane that passes through the origin. Since $\overrightarrow{H_{hkl}}$ is perpendicular to the plane (hkl), and passes through the origin, d_{hkl} is simply the magnitude of the projection of any of the vectors $\frac{\overrightarrow{a_1}}{h}$, $\frac{\overrightarrow{a_2}}{k}$, or $\frac{\overrightarrow{a_3}}{l}$ along the direction of $\overrightarrow{H_{hkl}}$, or along $\hat{n}$

Therefore,

$$d_{hkl} = \frac{\overrightarrow{a_1}}{h}.\hat{n} = \frac{\overrightarrow{a_1}}{h}.\frac{\left(h\overrightarrow{b_1} + k\overrightarrow{b_2} + l\overrightarrow{b_3}\right)}{\left|\overrightarrow{H_{hkl}}\right|} = \frac{1}{\left|\overrightarrow{H_{hkl}}\right|}$$

We started this discussion with merely an enforcement that the integers $h, k,$ and l, which were miller indices of a plane in real space, matched the components of a vector in reciprocal space along its axes. However we now find that due to the manner in which reciprocal space has been defined, the real space plane (hkl), and the reciprocal lattice vector $\overrightarrow{H_{hkl}}$, are related in interesting ways. Specifically, we find that:

1) $\overrightarrow{H_{hkl}}$ is perpendicular to the plane (hkl)
2) $d_{hkl} = \frac{1}{\left|\overrightarrow{H_{hkl}}\right|}$

In the next class we will look at the description of diffraction, in the context of the reciprocal space.